

Performing a medical examination assessing the conduction of nerves, a physician sends an electric impulse down the forearm of a patient. What is the average velocity of the electric signal if it arrives at an electrode 25 cm away from its initial position following 5 ms of travel?

- ☐ A. 0.2 m/s [3%]
- ☐ B. 5 m/s [16%]
- ☒ C. 50 m/s [78%]
- ☐ D. 200 m/s [2%]

Correct

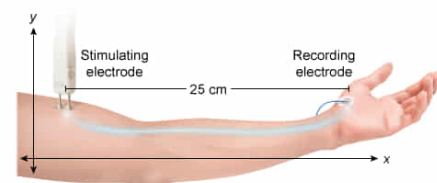
78% answered correctly

Explanation:

Average velocity from time & distance

$$v = \frac{\Delta x}{t} = \frac{x_f - x_i}{t}$$

* v Velocity
* $\Delta x, x_f - x_i$ Displacement
* t Time



In translational motion, **displacement** refers to the difference between the final and initial position of an object. Unlike distance, displacement is a **vector quantity** that contains both magnitude and direction. Displacement is also **path-independent** and can be determined by knowing only the initial and final position of an object (ie, the path along which an object travels is irrelevant).

Velocity (v), a vector quantity that expresses displacement per unit time, can be assessed instantaneously or as an average. Instantaneous velocity refers to the momentary velocity of an object at a single point in time. Alternatively, **average velocity** refers to the mean velocity of an object over a *period* of time. The motion of an object over a period of time can be expressed as an average velocity on an x,y -coordinate system by dividing the x - and y -components of displacement ($x_f - x_i$ and $y_f - y_i$, respectively) by the interval of **time (t)** for that motion:

$$v_x = \frac{x_f - x_i}{t}$$

In this question, a medical practitioner investigates the function of nerves by eliciting the flow of electrons down a patient's nerve. Although the electrons follow a winding course through the forearm, the total displacement of the electrons during the 5 ms journey is 25 cm. By selecting an x,y -coordinate system oriented such that the motion of the electrons takes place along the x -axis exclusively (no y -axis displacement, $v_y = 0$), only the x -component of average velocity v_x must be considered.

Substitution of displacement and time values into the equation for displacement yields:

$$v_x = \frac{(25 \text{ cm})}{5 \text{ ms}}$$

Converting all values to **standard units** results in:

$$v_x = \frac{25 \text{ cm}}{5 \text{ ms}} = 50 \text{ m/s}$$

Therefore, it can be concluded that the average velocity of electrons that travel down the patient's arm is 50 m/s.

Educational objective:

The average velocity of an object undergoing translational motion can be calculated as the total displacement of that object divided by the interval of time for that motion. Displacement and velocity are both vector quantities.